METRIC

MIL-DTL-32666

6 October 2020

DETAIL SPECIFICATION

PENETRATORS, PIPING, MECHANICAL, SURFACE SHIP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanical bulkhead and deck piping transit penetrators for use on Naval surface ships, using pipe specified in S9086-RK-STM-010/505, with a normal operating temperature of -20 to 80 °C (-4 to 175 °F).

1.2 Classification. Penetrators are classified in accordance with 1.3.1 through 1.3.4.

1.3 Part or identifying number (PIN). PINs to be used for penetrators acquired to this specification are created as follows:

M	**32666**	-	**X**	**XXX**	**X**	-	**XXXX/XXXX(X)**
Prefix for Military Specification	Specification Number	-	Item Type (see 1.3.1)	Notional Frame Size (R and P items only) (see 1.3.1 and 1.3.2)	Item Material (see 1.3.3)	-	Inserts (see 1.3.4)

1.3.1 Item type. Items are designated by a single letter specified as follows:

A = Assembly, end-packing, rectangular frame, square blocks (see 6.4.3)

B = Piping insert blocks (see 6.4.8) or blanking blocks (see 6.4.1)

I = Filler inserts (see 6.4.4)

R or P = Frame, round (R) or rectangular (P) (see 1.3.2 and 6.4.5)

S = Stay plates (see 6.4.11)

C = Compression plates (see 6.4.2)

E = Assembly, inserts, round frame, square blocks

H = Assembly, inserts, round frame, single piping

K = Assembly, inserts, round frame, single piping with adjustable outer diameter (peel-away)

1.3.2 Frame size. Frame size designation is a three-numerical-digit maximum Diameter Nominal (DN) pipe size for either round or rectangular frames as noted in table I. For non-frame items, the PIN code is "---". "Frame" in this context refers to the external enclosing material of a piping penetrator assembly. A piping penetrator frame is not intended to provide structural compensation for holes cut into a ship's structural bulkheads, stiffeners, supports, or any other structural members.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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TABLE I. Mechanical penetrators, pipes, and frames.

NPS	DN Size	Outside Diameter mm (inches)	Similar MIL-DTL-24705/1 block	Frame (see 1.3.1 and 1.3.2)
¼	008	13.72 (0.540)	BN2013 BN3013	R or P
1	025	33.4 (1.315)	BN4033 BN3033	R or P
1½	040	48.26 (1.9)	BN6048	R or P
2	050	60.33 (2.375)	BN8060	R or P
3	080	88.9 (3.5)	BN12087	R or P
4	100	114.3 (4.5)	none	R or P
6	150	168.28 (6.625)	none	R

1.3.3 Item material. Item material is as specified (see 3.3) with the item material designated by a single letter specified as follows:

A = Aluminum

B = Brass

C = Corrosion-resistant steel (CRES)

E = Elastomer or non-metallic

N = Not specified

S = Steel

Z = Galvanized steel

1.3.4 Piping insert blocks. Piping insert blocks are differentiated by an item designation consisting of the following format, “xxxx/xx(x)”. The first two digits indicate the exterior width of the piping insert block in millimeters. The next two digits indicate the diameter of the passage of the piping insert block in millimeters. If the piping passage diameter is fixed, the remaining digits will not be used. If the block has an adjustable piping passage diameter, the next four digits indicate the minimum and maximum diameters in millimeters. When an insert block is designed to accept multiple pipes, the number of pipes that can be accommodated will be included in the parentheses “( )” at the end of the item designation. Similar inserts are used in MIL-DTL-24705 cable penetrators, and some examples are provided in table I.

1.3.5 Example PINs. Illustrative PIN construction examples are provided below.

a. PIN M32666-P080S-8060(4) is a rectangular steel frame penetrator, notional DN size 80, which has blocking for 4 DN size 50 pipes.

b. PIN M32666-B---E-4030/50 is an elastomer insert of adjustable diameter of 30 to 50 millimeters.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

C-T-91 - Tallow, Inedible

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

FED-STD-H28/2 - Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads–UN and UNR Thread Forms

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for

MIL-DTL-16884 - Fuel, Naval Distillate

MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service

MIL-DTL-24705/1 - Penetrators, Multiple Cable, Electric Cable, Rectangular Metal Frame, Square-Faced Blocks

MIL-DTL-24705/3 - Penetrators, Multiple Cable, Electric Cable, Round Frame, Square-Faced Blocks

MIL-PRF-87252 - Coolant Fluid, Hydrolytically Stable, Dielectric

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-889 - Dissimilar Metals

MIL-STD-3020 - Fire Resistance of U.S. Naval Surface Ships

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Technical Data Management Information System [TDMIS] at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering [hard copy] via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASME INTERNATIONAL

ASME B18.2.1 - Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

ASME B18.2.2 - Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A106/A106M - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service

ASTM A108 - Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished

ASTM A131/A131M - Standard Specification for Structural Steel for Ships

ASTM A276/A276M - Standard Specification for Stainless Steel Bars and Shapes

ASTM A312/A312M - Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes

ASTM A342/A342M - Standard Test Methods for Permeability of Weakly Magnetic Materials

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ASTM A967/A967M	-	Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts
ASTM A1085/A1085M	-	Standard Specification for Cold-Formed Welded Carbon Steel Hollow Structural Sections (HSS)
ASTM B21/B21M	-	Standard Specification for Naval Brass Rod, Bar, and Shapes
ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B209	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B221	-	Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM D395	-	Standard Test Methods for Rubber Property—Compression Set
ASTM D2240	-	Standard Test Method for Rubber Property—Durometer Hardness

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10012	-	Measurement Management Systems — Requirements for Measurement Processes and Measuring Equipment
ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available online at www.iso.org.)

SAE INTERNATIONAL

SAE AMS3249	-	Ethylene Propylene (EPDM) Rubber, Hydrazine-Base-Fluid Resistant 75 - 85
SAE J1086	-	Numbering Metals and Alloys

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 (see 6.3).

3.2 Typical configurations. Typical round and rectangular piping penetrator configurations are shown on figures 1 and 2.

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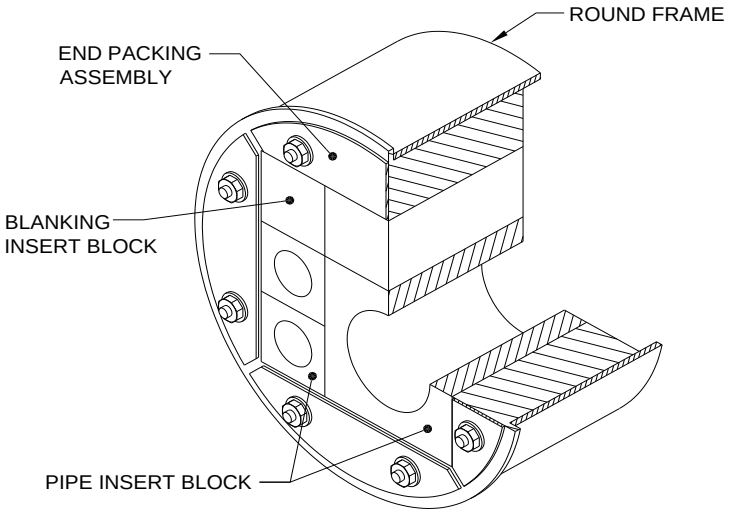


FIGURE 1. Typical round frame multi-pipe penetrator configuration (note cutaway).

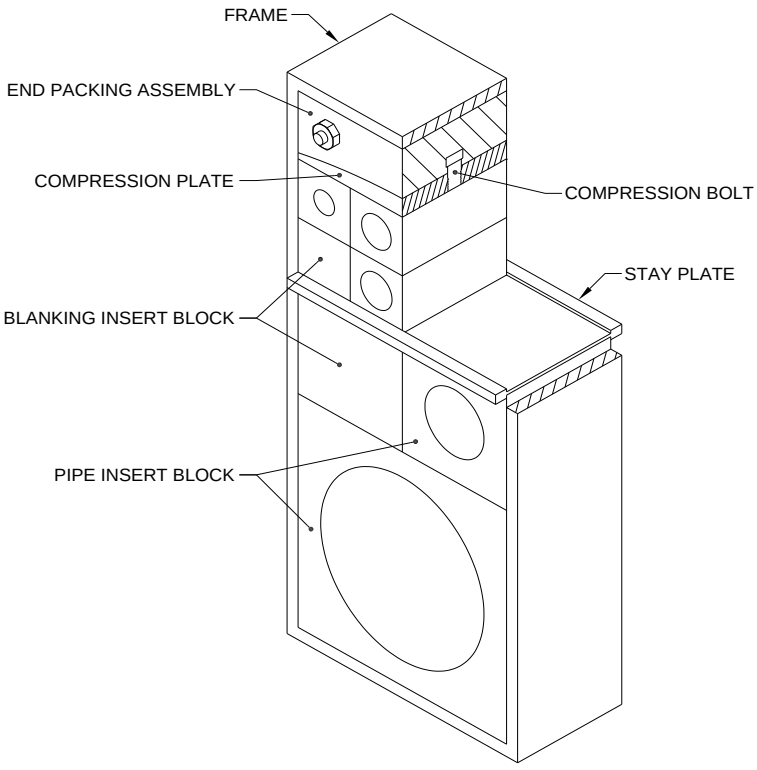


FIGURE 2. Typical rectangular frame multi-pipe penetrator configuration (note cutaway).

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3.3 Materials. Materials shall be as specified herein. When not specified, material may be selected that will satisfactorily perform the intended function of the component in the equipment assembly, provided the component meets the requirements of the specification.

3.3.1 Metals and alloys. Metals and alloys for frames and accessories shall be corrosion resistant or be given a corrosion-resistant treatment or coating where non-corrosion-resistant steel is used for a frame that is to be welded to a bulkhead or other partition. The metal shall be cleaned and receive a gray paint primer. Metals and alloys not covered in this specification shall be approved by the Naval Sea Systems Command (NAVSEA) before use in a piping penetrator.

3.3.1.1 Corrosion-resisting and nonmagnetic metals. When nonmagnetic material or resistance to corrosion is required, the material shall be selected from the following:

a. Alloys, aluminum: types UNS A95052[1] or A95086 in accordance with ASTM B221 or A95083 in accordance with ASTM B209.

b. Brass: type C46400 in accordance with ASTM B21/B21M.

c. CRES: ASTM A276/A276M grades UNS S30403 (304L), S30603 (316L), S32100 (321), or S34700 (347); ASTM A276/A276M condition A; and passivated in accordance with ASTM A967/A967M. S30403 shall not be used in potentially saltwater-wetted areas.

3.3.1.2 Steel. Steels used in penetrator structures shall be grades A, B, or C in accordance with ASTM A131/A131M or grade C1018 in accordance with ASTM A108.

3.3.1.3 Dissimilar metals. Dissimilar metals shall be in accordance with MIL-STD-889.

3.3.2 Non-metallic materials.

3.3.2.1 Glass-reinforced plastic (GRP). GRP shall not be used for piping penetrator frames.

3.3.2.2 Insert material items. Insert material items (see 6.4.6) shall be constructed of material that meets the requirements specified herein. Insert materials in use are Lycron™, Roxylon™, SAE AMS3249 ethylene propylene 75 to 85, and others; all insert materials shall be shown to meet all requirements of this specification prior to use.

3.3.2.3 Lubricant. Lubricant (see 6.4.7) shall be used to assure even compression throughout transit during assembly. Assembly lubricant shall be nontoxic and in accordance with C-T-91 or a polyethylene glycol gel.

3.3.3 Fasteners. Unless otherwise specified (see 6.2), fasteners shall be in accordance with MIL-DTL-1222. Bolts and nuts shall be in accordance with ASME B18.2.1 and ASME B18.2.2.

3.3.4 Prohibited materials. The penetrators shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.3.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Construction.

3.4.1 General. The piping penetrator frame, end-packing assembly (see 6.4.3), compression plate (see 6.4.2), stay plates, piping insert blocks, blanking insert blocks (see 6.4.1), and filler inserts (see 6.4.4) shall be of the design, construction, and physical characteristics as specified below (see figures 3, 4, and 5). Piping insert blocks may be either rectangular-faced (see 6.4.9) or square-faced (see 6.4.12).

[1] Unified Numbering System (UNS) designations are in accordance with SAE J1086.

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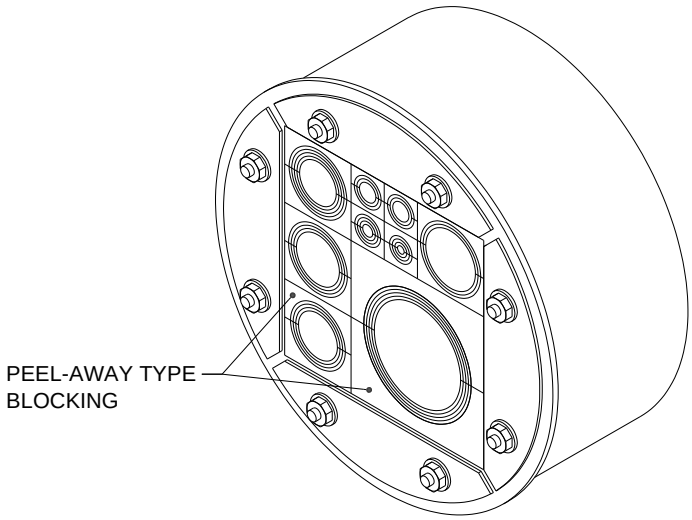


FIGURE 3. Generic round multiple-pipe, peel-away type insert blocking (frame omitted).

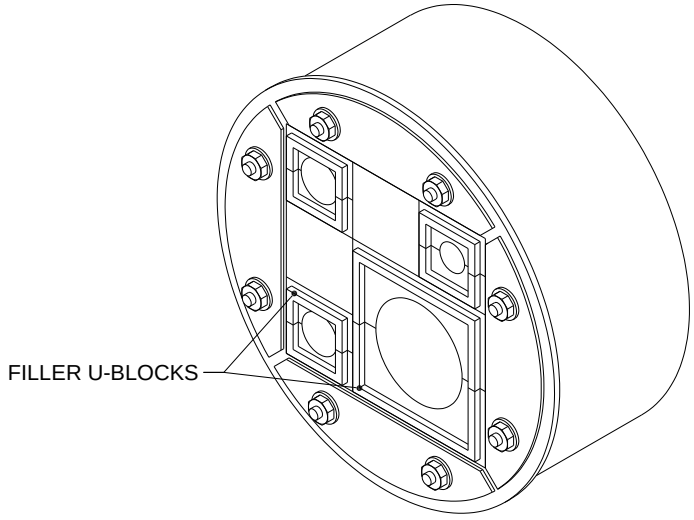


FIGURE 4. Generic round multiple-pipe, filler insert blocks (frame omitted).

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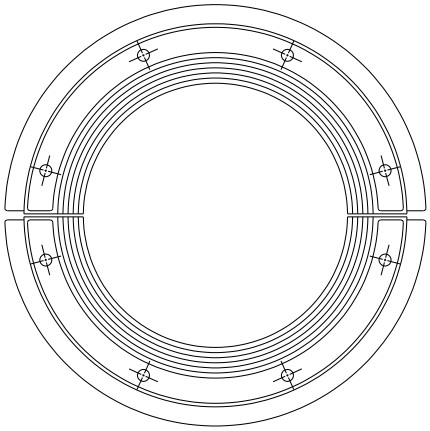


FIGURE 5. Generic round penetrator, single-pipe, peel-away type insert (frame omitted).

3.4.1.1 Round penetrators. Round frame penetrators may serve multiple or single pipes. They shall be furnished with metal compression plates covering the unused perimeter around the square or round pipe transit area.

a. Dimensions. Round penetrator frames shall be as specified (see 6.2) in the range in diameter from 50 to 240 millimeters (2 to 10 inches) and accommodate the pipe sizes noted in table I in single- or multiple-pipe configurations. The depth of the frame shall be as specified (see 6.2), in the range of 60 to 305 millimeters (2½ to 12 inches).

b. Wall thickness. Wall thickness shall, at minimum, meet American National Standard Schedule 40 pipe dimensions equal to the frame diameter up to a maximum wall thickness in accordance with the ASTM A1085/A1085M Hollow Structural Sections (HSS) shape database.

c. Insert and blanking blocks. Peel-away or sized inserts as specified (see 6.2) shall be manufactured to accommodate the pipe sizes of table I. Blanking blocks shall be manufactured to allow fitting of insert blocks into a given frame with no gaps in coverage.

3.4.1.2 Rectangular penetrators. Rectangular frame penetrators may serve multiple or single pipes. Pipes shall be positioned such that similar sizes lie in the same plane of the penetrator, and planes shall be separated by metal stay plates. See figures 6 and 7.

a. Dimensions. Rectangular frame penetrator frames shall be as specified (see 6.2) in the range of 100 to 300 millimeters in height, 120 to 150 millimeters in width, and 60 millimeters in depth; they shall accommodate the pipe sizes noted in table I in single- or multiple-pipe configurations.

b. Wall thickness. Wall thickness of rectangular frames shall be a minimum of 10 millimeters or the greater thickness of the related ASTM A1085/A1085M HSS shape.

c. Compression plates and end-packing. Compression plates and end-packing shall be sized to match the frame and inserts of a rectangular penetrator. End-packing shall be in configurations in accordance with MIL-DTL-24705/1 or another configuration approved by NAVSEA. Compression plates shall be provided with a means to restrain movement of the compression plate since compression takes place during assembly.

d. Insert and blanking blocks. Peel-away or sized inserts as specified (see 6.2) shall be manufactured to accommodate the pipe sizes of table I. Blanking blocks shall be manufactured to allow fitting of insert blocks into a given frame with no gaps in coverage.

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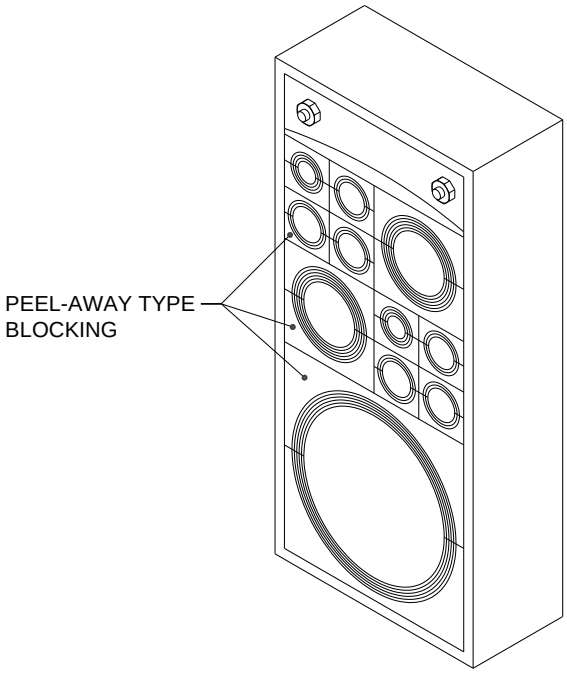


FIGURE 6. Generic rectangular multiple-pipe, peel-away type blocking.

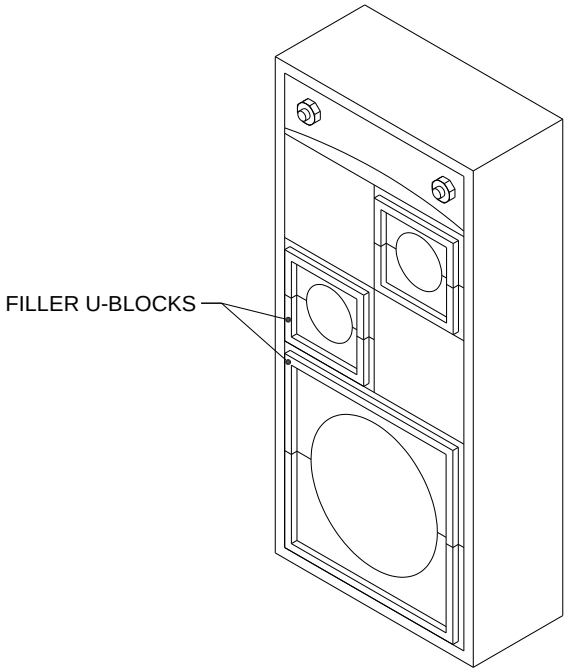


FIGURE 7. Generic rectangular multiple-pipe, filler insert blocks.

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3.4.2 Self-tapping hardware. Self-tapping screws or other self-tapping devices shall not be used in any assembled piping penetrator.

3.4.3 Threads. Threads shall be unified form class 2A or 2B in accordance with FED-STD-H28 and FED-STD-H28/2.

3.4.4 Castings. Cast components shall be free from cold shuts, blow holes, or cracks that may in any way affect the strength or serviceability of the casting. Fins or burrs on cast surfaces shall be removed.

3.4.5 Forming and machining operations. Forming and machining operations shall be completed before any plating or finish is applied.

3.4.6 Sharp edges. There shall be no sharp edges.

3.4.7 Welding. Welding shall be in accordance with S9074-AR-GIB-010/278.

3.4.8 Molded parts. Molded parts shall be free of flash, blow holes, or cracks. Small gas bubbles that form on the face and top surface shall not be cause for rejection, provided the molded parts are in accordance with the requirements specified herein and performance is unaffected. Flash shall be removed and the width of the flash edges shall not be greater than 0.8 millimeter (1/32 inch). Measures shall be taken in molding to ensure that stress buildup does not occur; or, the items shall be satisfactorily treated to relieve these stresses to prevent deterioration or failure of a part or assembly. The stress-relieving process shall be as specified by the manufacturer of the raw materials.

3.4.9 Surface finish for metals. A maximum surface roughness of 6.35 micrometers (250 microinches) shall be permissible for machined metal parts. Surface finish shall be in accordance with ASME B46.1.

3.4.10 Nonmagnetic penetrations. When specified to be nonmagnetic (see 6.2), penetrations shall meet the nonmagnetic requirements of 3.5.4.

3.4.11 Special tools and equipment. Unless approved by NAVSEA, special tools (see 6.4.10) and equipment for installation, removal, or replacement of frames or for reconfiguration of accessories and insert blocks shall not be required.

3.5 Performance. Piping penetrator frames, accessories, and insert material items (see 6.4.6) shall be constructed to meet the performance requirements specified herein.

3.5.1 Vibration. Piping penetrators shall withstand the vibration test of 4.4.3.

3.5.2 High-impact (H.I.) shock. Piping penetrators shall withstand the shock test of 4.4.4.

3.5.3 Watertight. Piping penetrators shall be watertight and shall exhibit leakage not greater than specified in 4.4.5.

3.5.4 Nonmagnetic. When nonmagnetic piping penetrators are specified (see 3.4.10 and 6.2), the metal parts shall indicate a permeability level of less than 2 henries per meter (see 4.4.6).

3.5.5 Salt spray. Piping penetrators shall exhibit no corrosion or physical damage when subject to the testing of 4.4.7.

3.5.6 Fire resistance. Piping penetrators shall withstand the fire resistance test of 4.4.8.

3.5.7 Durometer. Insert material shall have a class A durometer hardness of 81±6 when tested in accordance with 4.4.9.

3.5.8 Compression. Insert material shall have a compression not greater than 2 percent (see 4.4.10).

3.5.9 Expansion. Insert material shall have an expansion not greater than 1 percent (see 4.4.11).

3.5.10 Fluid immersion. Insert material shall exhibit no apparent physical change, and the durometer hardness shall not exceed ±5 percent change (see 4.4.12).

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3.6 Marking. Marking shall be as specified in 3.6.1 and 3.6.2.

3.6.1 Frames and accessories. Accessories consist of end-packing assemblies, compression plates, and stay plates (see 6.4). Unless otherwise specified (see 6.2), frames and accessories shall be permanently marked by ink, stamping, molding, or casting or be engraved with the following:

a. Military part number.

b. Manufacturer's name or logo.

Identification marking shall be at a location that will be clearly visible once the piping penetrator has been assembled. Printed characters shall be 3 millimeters high or larger and shall be clear, complete, and legible to the naked eye.

3.6.2 Piping insert blocks. Piping insert blocks shall be permanently marked by molding, engraving, or indelible stamping with the following:

a. Exterior width and passage diameter in millimeters.

b. Passage diameter in inches.

c. Manufacturer's name or logo.

Unless otherwise specified (see 6.2), passage diameter shall be marked in millimeters and inches on each end face of the piping insert block. Location of the marking required for small block sizes or for blocks with large passage diameters that restrict marking space availability or other required identification marking shall be at the option of the manufacturer. Printed characters shall be complete and legible to the naked eye.

3.7 Workmanship. Penetrator components shall be thoroughly cleaned both inside and out and shall be free from burrs, loose sand, grit, scale, metal shavings, and all other contaminants that can affect performance and damage sealing surfaces. There shall be no sharp edges. All sealing, rubbing, and coated surfaces shall be smooth with no blemishes, scratches, scrapes, or gouges.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.1.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the following conditions:

a. Temperature: -20 to 80 °C (-4 to 175 °F), except for seawater tests, which may be -2 to 80 °C (28 to 175 °F).

b. Atmospheric pressure: 550 to 800 millimeters of mercury (10.6 to 15.5 pounds per square inch [psi]).

c. Relative humidity: 20 to 80 percent.

4.1.2 Instrumentation calibration. Measuring and test equipment used in the performance of tests shall be calibrated in accordance with ISO 10012.

4.2 First article inspection. First article inspection shall be performed on a test specimen of each frame type of mechanical bulkhead penetrator when a first article sample is required (see 3.1). This inspection shall include the examinations and tests of table II.

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TABLE II. First article and conformance verifications.

Inspection	Requirement	Test method	First article	Conformance
Visual and dimensional examination	3.4 and 3.6	4.4.2	X	X
Materials	3.3	4.5	X	
Prohibited materials	3.3.4	4.6	X	
Vibration	3.5.1	4.4.3	X	
Shock	3.5.2	4.4.4	X	
Watertightness	3.5.3	4.4.5	X	
Nonmagnetic 1/	3.5.4	4.4.6	X	X
Salt spray	3.5.5	4.4.7	X	
Fire resistance	3.5.6	4.4.8	X	
Durometer 2/	3.5.7	4.4.9	X	X
Compression 2/	3.5.8	4.4.10	X	X
Expansion 2/	3.5.9	4.4.11	X	
Fluid immersion 2/	3.5.10	4.4.12	X	

NOTES:

1/ Shall be conducted when specified (see 6.2) on penetrator assemblies having aluminum and CRES frames only.

2/ Shall be conducted on penetrator insert material separately from penetrator assemblies.

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests of table II.

4.3.1 Samples. The samples to be tested shall be produced with the same processes, materials, and equipment normally used in production.

4.3.1.1 Sampling for examinations and tests. A random sample of penetrators shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.3.1.2 Lot. A lot shall consist of all penetrator assemblies, frames, or inserts produced under the same conditions and offered for delivery at one time, as specified (see 6.2). A sample penetrator assembly shall consist of one complete test specimen in accordance with 4.3.1.1.

4.3.2 Piping penetrator test specimens. Piping penetrator test specimens (see 3.1) of each frame material, shall be tested as specified in table II. If a manufacturer produces only insert material items (for assembly in a frame produced by others), they shall be assembled in a frame and tested as specified in table II. The test specimen shall consist of a piping penetrator frame, as well as the appropriate insert material items and accessories, loaded with pipes in accordance with ASTM A106/A106M or ASTM A312/A312M (Schedule 10 or greater) as shown on figures 8, 9, and 10. The specified pipes shall be 1 meter in length and supported externally to the penetrator assembly. Using the test stand specified in 4.4.1 or an approved equivalent, the pipes shall be routed through the penetrator frame so that 0.75 meter of piping shall extend from the side of the frame that will not be directly exposed to the fire or water during testing.

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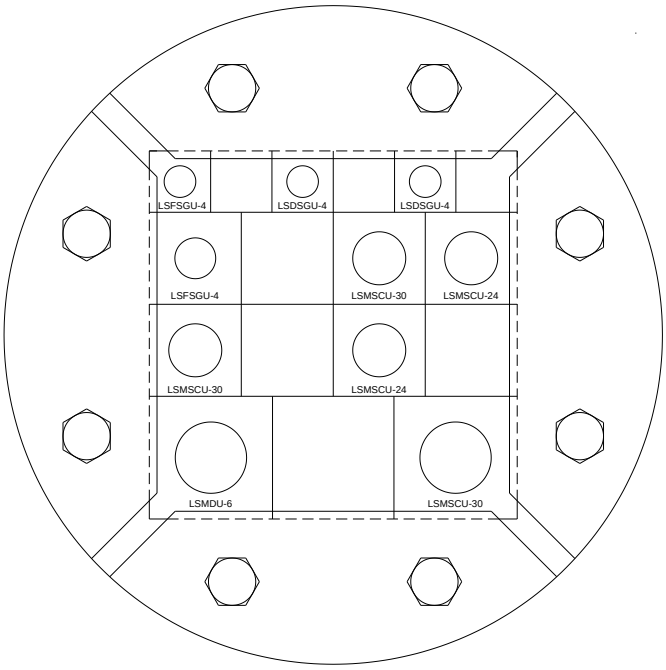


FIGURE 8. Generic test configuration round multiple (frame omitted) (see description and figure in MIL-DTL-24705/3 [RGP 8 size]).

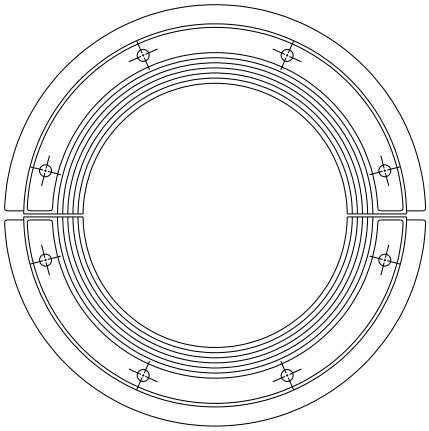


FIGURE 9. Generic test configuration, round single-pipe.

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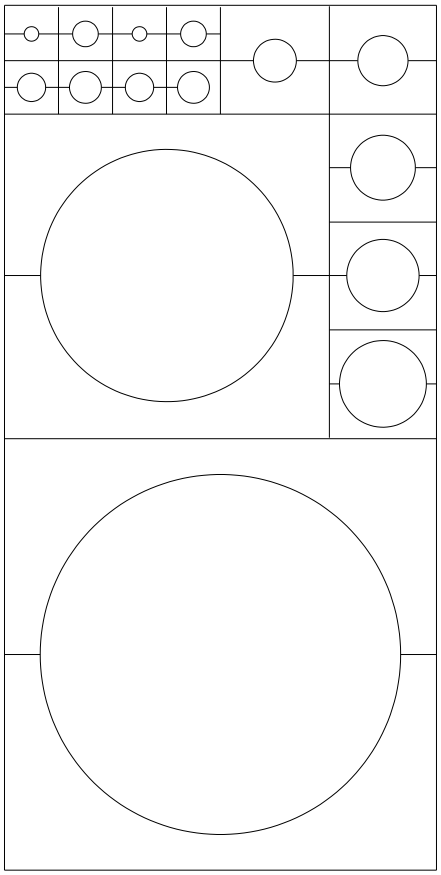


FIGURE 10. Generic test configuration, rectangular multiple-pipe (frame omitted) (see description and figure in MIL-DTL-24705/1 [RGS 8 size]).

4.3.3 Insert material sample. Separate insert material samples shall be tested as specified in table II.

4.3.4 Failure. Failure of any sample to meet the requirements specified herein shall be cause for rejection.

4.4 Methods of inspection.

4.4.1 Test stand assembly. The test stand assembly used to test the piping penetrator test specimen shall be in accordance with the MIL-STD-3020 appendices on pipe and duct penetration test configuration and cable transit test configuration or shall be approved by NAVSEA (see 4.3.2).

4.4.2 Visual and dimensional examination. Piping penetrator frames, accessories, and insert material items shall be visually and dimensionally examined to verify that the design, construction, physical dimensions, and marking are as specified in the applicable requirements (see 3.4 and 3.6).

4.4.3 Vibration. The piping penetrator test specimen shall be subjected to the type I vibration test in accordance with MIL-STD-167-1. Following the test, the piping penetrator test specimen shall meet the acceptability requirements of MIL-STD-167-1 and shall pass the H.I. shock test specified in 4.4.4 and the watertightness test specified in 4.4.5, in that order.

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4.4.4 Shock The piping penetrator test specimen shall be subjected to a grade A, class I, type A shock test in accordance with MIL-DTL-901. Standard lightweight and medium-weight shock machine fixtures shall be used in accordance with MIL-DTL-901. Following the test, the piping penetrator shall meet the acceptability requirements of MIL-DTL-901 and shall pass the watertightness test specified in 4.4.5.

4.4.5 Watertightness. The piping penetrator test specimen mounted on the test plate shall be bolted to the hydro chamber so that one side of the test specimen is exposed to water and the opposite side of the test specimen is exposed to air. The hydro chamber shall then be filled with water and pressurized to 1.9 bars (27.5 psi). This pressure shall be maintained for a minimum of 10 minutes, unless specimen failure occurs sooner. The test specimen shall be considered as having failed the watertightness test if, after 10 minutes, the water pressure has decreased to less than 1.9 bars or if any visually discernable leakage has occurred.

4.4.6 Nonmagnetic. When applicable (see 3.4.10 and 3.5.4), nonmagnetic parts shall be tested with a permeability indicator in accordance with ASTM A342/A342M. Nonmagnetic parts shall meet the requirements specified in 3.5.4.

4.4.7 Salt spray test. The piping penetrator test specimen shall be tested in accordance with ASTM B117. The exposure time shall be 48 hours. Failure to meet the criteria of 3.5.5 shall be cause for rejection.

4.4.8 Fire resistance test. The piping penetrator test specimen shall be as specified in 4.3.2. A fire resistance and temperature rise test shall be conducted in accordance with the requirements for class N-30 for unrestricted applications in MIL-STD-3020. This testing shall be conducted after the tests of 4.4.3, 4.4.4, and 4.4.5 on the same test specimen. Failure to meet class N-30 criteria shall be cause for rejection.

4.4.9 Durometer. A sample of each insert material shall be subjected to a type A durometer test in accordance with ASTM D2240. Insert materials shall meet the requirements specified in 3.5.7.

4.4.10 Compression. A sample of each insert material shall be tested in accordance with ASTM D395. Insert materials shall meet the requirements specified in 3.5.8.

4.4.11 Expansion. A sample of the insert material shall be placed into an oven with the temperature held constant at 80 to 85 °C (176 to 185 °F) for not less than 1 hour. Before the test, the sample shall be measured for its actual dimensions. After the test has concluded and the sample has been cooled to ambient temperature, the sample dimensions shall be measured. The percentage change in volume shall be calculated as follows:

$$\text{Expansion percentage} = \frac{(\text{Volume after test}) - (\text{Volume before test})}{(\text{Volume before test})} \times 100$$

This test shall be repeated three times sequentially to indicate cyclic consistency. At all measurement points, the insert material shall meet the requirements specified in 3.5.9.

4.4.12 Fluid immersion. A sample insert material shall be placed face down in a container containing 13 millimeters (½ inch) of fluid for not less than 2 hours. The fluids and temperatures shall be as specified in table III. The samples shall then be removed, blotted to remove excess fluid, and suspended in air at room temperature for 3½ to 4½ hours. Each sample shall then be tested again as specified in 4.4.9. The test shall be repeated for each specified fluid. Insert material shall meet the durometer requirements specified in 3.5.10.

TABLE III. Test fluids and temperatures.

Fluid	Test temperatures
Fuel oil (in accordance with MIL-DTL-16884)	See 4.1.1
Hydraulic fluid (in accordance with MIL-PRF-17331)	See 4.1.1
Dielectric heat transfer fluid (in accordance with MIL-PRF-87252)	See 4.1.1
Seawater	See 4.1.1

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4.5 Materials. Materials verification shall be as specified (see 3.3 and 6.2).

4.6 Prohibited materials. Prohibited materials shall be verified as specified (see 3.3.4 and 6.2).

4.7 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to the Technical Authority (see 6.4.13). Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or the International Code Council's International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Piping penetrators are intended for use in maintaining watertight, airtight, fumetight, and firetight integrity of piping penetrations through decks and bulkheads (except those of tank boundaries) of Naval surface ships. Piping penetrators herein meet military shock and other criteria in excess of civilian classification society requirements and are intended to be used primarily with pipe specified in S9086-RK-STM-010/505. Penetrator assemblies that meet the requirements specified herein are expected to be highly reliable for the service life of a Navy surface ship or aircraft carrier.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Part or identifying number (see 1.3).
c. When first article inspection is required (see 3.1).
d. Requirement for materials verification (see 3.3 and 4.5).
e. Fasteners, if other than specified (see 3.3.3).
f. Requirement for prohibited materials verification (see 3.3.4 and 4.6).
g. Dimensions of round penetrators (see 3.4.1.1.a).
h. Insert type for round penetrators (see 3.4.1.1.c).
i. Dimensions of rectangular penetrators (see 3.4.1.2.a).
j. Insert type for rectangular penetrators (see 3.4.1.2.d).
k. Nonmagnetic requirements, if any (see 3.4.10, 3.5.4, and table II).
l. Marking for frames and accessories, if other than specified (see 3.6.1).
m. Passage diameter of piping insert blocks, if other than specified (see 3.6.2).
n. Inspection conditions, if other than specified (see 4.1.1).
o. Acceptance Quality Limit (AQL) of 1.0 (see 4.3.1.1).
p. Lot components (see 4.3.1.2).
q. Additional fire testing provisions (see 4.7 and 6.5).

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r. Packaging requirements (see 5.1).

s. Recommended documentation (see 6.6).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors regarding whether the item should be a preproduction sample, a first article sample, a first production item, samples selected from the first lot size of production items, or a standard production item from the contractor's current inventory, as well as the number of items to be tested. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government; they should also provide that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Blanking insert block. The fire blanking insert block is a solid piping penetrator insert block with no passages.

6.4.2 Compression plate. The compression plate applies compression on all internal parts and fills the void between the top row of insert blocks and the frame, used in conjunction with the end-packing assembly.

6.4.3 End-packing assembly. The end-packing compression assembly is a component consisting of insert material sandwiched between metallic plates. It is used in conjunction with a compression plate to apply compression on all internal parts and to fill the void between the top row of insert blocks and the frame. This may also be a wedge, which is a compression block used in rectangular frames without built-in compression.

6.4.4 Filler inserts. Filler inserts are rectangular or specially shaped components provided in multiple units or single pieces that are used to fill void spaces in piping penetrator assemblies.

6.4.5 Frame. The frame is the housing into which accessories and insert material items are assembled. When used to pass pipe through bulkheads or decks, the piping penetrator frame is welded to the bulkhead or deck. A piping penetrator frame is not intended to provide structural compensation for holes cut into a ship's structural bulkheads, stiffeners, supports or any other structural members.

6.4.6 Insert material. Insert material consists of piping insert block, blanking insert block, and filler inserts.

6.4.7 Lubricant. With regard to piping penetrators, lubricant is a substance that allows for ease of initial installation of insert material and piping during assembly.

6.4.8 Piping insert block. The piping insert block is a one- or two-piece block with single or multiple passages to accommodate a pipe or pipes. The block fits around the pipe or pipes to form an airtight, fumetight, watertight, and firetight seal when installed in a piping penetrator assembly. This may be sized exactly for certain sizes or composed of layers that may be peeled off for tailoring

6.4.9 Rectangular-faced block. The rectangular-faced block is a two-piece piping insert block that has a rectangular-shaped face with round or half-round passages.

6.4.10 Special tools. Special tools are defined as tools not listed in the GSA Global Supply Catalog.

6.4.11 Stay plates. Stay plates are components that align piping insert blocks, blanking insert blocks, and filler inserts in position and assist in providing even compression throughout the transit when installed in a piping penetrator assembly.

6.4.12 Square-faced block. The square-faced block is a one- or two-piece piping insert block that has a square-shaped face with a single passage.

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6.4.13 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.5 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.6 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition (see 6.2) of any mechanical bulkhead penetrators:

a. Engineering drawings.
b. Materials listing.
c. Installation drawings.
d. Technical manual in accordance with MIL-DTL-24784.
e. Visual aids and maintenance cards in accordance with MIL-STD-3034.
f. Test reports.
g. Certifications.
h. Onboard repair parts list.

6.7 Subject term (key word) listing.

Fire
Frames
Inserts
Stay plates

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH
Air Force – 03

Preparing Activity:
Navy – SH
(Project 4730-2020-057)

Review activities:
Army – AV, CR
Air Force – 11, 70, 71
DLA – CC

Civil agency:
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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